25

Lung

(Carcinoid tumors are included. Sarcomas and other rare tumors are not included.)

At-A-Glance

SUMMARY OF CHANGES

- This staging system is now recommended for the classification of both non-small cell and small cell lung carcinomas and for carcinoid tumors of the lung
- The T classifications have been redefined:
  - T1 has been subclassified into T1a (≤2 cm in size) and T1b (>2–3 cm in size)
  - T2 has been subclassified into T2a (>3–5 cm in size) and T2b (>5–7 cm in size)
  - T2 (>7 cm in size) has been reclassified as T3
  - Multiple tumor nodules in the same lobe have been reclassified from T4 to T3
  - Multiple tumor nodules in the same lung but a different lobe have been reclassified from M1 to T4
- No changes have been made to the N classification. However, a new international lymph node map defining the anatomical boundaries for lymph node stations has been developed
- The M classifications have been redefined:
  - M1 has been subdivided into M1a and M1b
  - Malignant pleural and pericardial effusions have been reclassified from T4 to M1a
  - Separate tumor nodules in the contralateral lung are considered M1a
  - M1b designates distant metastases

ANATOMIC STAGE/PROGNOSTIC GROUPS

Occult Carcinoma	TX	N0	M0
Stage 0	Tis	N0	M0
Stage IA	T1a	N0	M0
	T1b	N0	M0
Stage IB	T2a	N0	M0
Stage IIA	T2b	N0	M0
	T1a	N1	M0
	T1b	N1	M0
	T2a	N1	M0
Stage IIB	T2b	N1	M0
	T3	N0	M0

ICD-O-3 TOPOGRAPHY CODES

C34.0 Main bronchus
C34.1 Upper lobe, lung
C34.2 Middle lobe, lung
C34.3 Lower lobe, lung
C34.8 Overlapping lesion of lung
C34.9 Lung, NOS

ANATOMIC STAGE/PROGNOSTIC GROUPS (*CONTINUED*)

Stage IIIA	T1a	N2	M0
	T1b	N2	M0
	T2a	N2	M0
	T2b	N2	M0
	T3	N1	M0
	T3	N2	M0
	T4	N0	M0
	T4	N1	M0
Stage IIIB	T1a	N3	M0
	T1b	N3	M0
	T2a	N3	M0
	T2b	N3	M0
	T3	N3	M0
	T4	N2	M0
	T4	N3	M0
Stage IV	Any T	Any N	M1a
	Any T	Any N	M1b

ICD-O-3 HISTOLOGY CODE RANGES

8000–8576, 8940–8950, 8980–8981

INTRODUCTION

Lung cancer is among the most common malignancies in the Western world and is the leading cause of cancer deaths in both men and women. The primary etiology of lung cancer is exposure to tobacco smoke. Other less common factors, such as asbestos exposure, may contribute to the development of lung cancer. In recent years, the level of tobacco exposure, generally expressed as the number of cigarette pack-years of smoking, has been correlated with the biology and clinical behavior of this malignancy. Lung cancer is usually diagnosed at an advanced stage and consequently the overall 5-year survival for patients is approximately 15%. However, patients diagnosed when the primary tumor is resectable experience 5-year survivals ranging from 20 to 80%. Clinical and pathologic staging is critical to selecting patients appropriately for surgery and multimodality therapy.

ANATOMY

Primary Site. Carcinomas of the lung arise either from the alveolar lining cells of the pulmonary parenchyma or from the mucosa of the tracheobronchial tree. The trachea, which lies in the middle mediastinum, divides into the right and left main bronchi, which extend into the right and left lungs, respectively. The bronchi then subdivide into the lobar bronchi in the upper, middle, and lower lobes on the right and the upper and lower lobes on the left. The lungs are encased in membranes called the visceral pleura. The inside of the chest cavity is lined by a similar membrane called the parietal pleura. The potential space between these two membranes is the pleural space. The mediastinum contains structures in between the lungs, including the heart, thymus, great vessels, lymph nodes, and esophagus.

The great vessels include:

Aorta
Superior vena cava
Inferior vena cava
Main pulmonary artery
Intrapericardial segments of the trunk of the right and left pulmonary artery
Intrapericardial segments of the superior and inferior right and left pulmonary veins

Regional Lymph Nodes. The regional lymph nodes extend from the supraclavicular region to the diaphragm. During the past three decades, two different lymph node maps have been used to describe the regional lymph nodes potentially involved by lung cancers. The first such map, proposed by Naruke (Figure 25.1) and officially endorsed by the Japan Lung Cancer Society, is used primarily in Japan. The second, the Mountain-Dresler modification of the American Thoracic Society (MD-ATS) lymph node map (Figure 25.2), is used in North America and Europe. The nomenclature for the anatomical locations of lymph nodes differs between these two maps especially with respect to nodes located in the paratracheal, tracheobronchial angle, and subcarinal areas. Recently, the International Association for the Study of Lung Cancer (IASLC) proposed a lymph node map (Figure 25.3) that reconciles the discrepancies between these two previous maps, considers other published proposals, and provides more detailed nomenclature for the anatomical boundaries of lymph nodes stations. Table 25.1 shows the definition for lymph node stations in all three maps. The IASLC lymph node map is now the recommended means

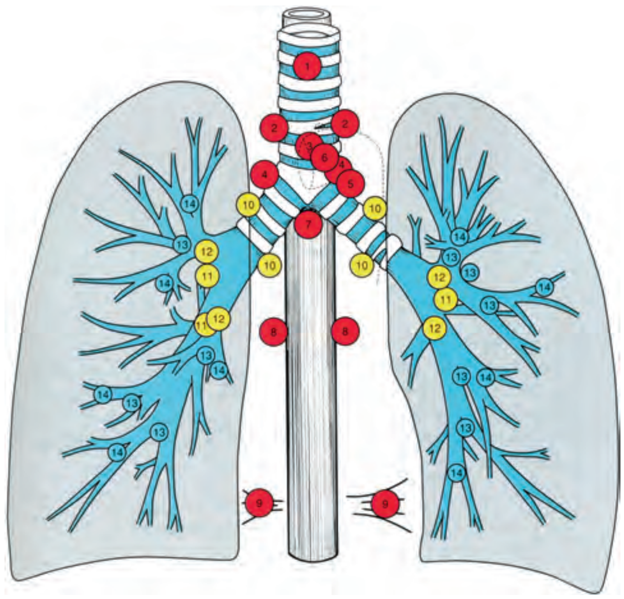

FIGURE 25.1. Naruke lymph node map. 1, Superior mediastinal or highest mediastinal; 2, paratracheal; 3, pretracheal: 3a, anterior mediastinal, 3p, retrotracheal or posterior mediastinal; 4, tracheobronchial; 5, subaortic or Botallo's; 6, paraaortic (ascending aorta); 7, subcarinal; 8, paraesophageal (below carina); 9, pulmonary ligament; 10, hilar; 11, interlobar; 12, lobar: upper lobe, middle lobe, and lower lobe; 13, segmental; 14, subsegmental. (From The Japan Lung Cancer Society. Classification of Lung Cancer. First English Edition. Tokyo: Kanehara & Co., 2000, used with permission.)

of describing regional lymph node involvement for lung cancers. Analyses of a large international lung cancer database suggest that for purposes of prognostic classification, it may be appropriate to amalgamate lymph node stations into "zones" (Figure 25.3). However, the use of lymph node "zones" for N staging remains investigational and needs to be confirmed by future prospective studies.

There are no evidence-based guidelines regarding the *number* of lymph nodes to be removed at surgery for adequate staging. However, adequate N staging is generally considered to include sampling or dissection of lymph nodes from stations 2R, 4R, 7, 10R, and 11R for right-sided tumors, and stations 5, 6, 7, 10 L, and 11 L for left-sided tumors. Station 9 lymph nodes should also be evaluated for lower lobe tumors. The more peripheral lymph nodes at stations 12–14 are usually evaluated by the pathologist in lobectomy or pneumonectomy specimens but may be separately removed when sublobar resections (e.g., segmentectomy) are performed. There is evidence to support the recommendation that histological examination of hilar and mediastinal lymphenectomy specimen(s) will ordinarily include 6 or more lymph nodes/stations. Three of these nodes/stations should be mediastinal, including the sub-carinal nodes and three from N1 nodes/stations.

Distant Metastatic Sites. The most common metastatic sites are the brain, bones, adrenal glands, contralateral lung, liver, pericardium, kidneys, and subcutaneous tissues. However, virtually any organ can be a site of metastatic disease.

RULES FOR CLASSIFICATION

Lung cancers are broadly classified as either non-small cell (approximately 85% of tumors) or small cell carcinomas (15% of tumors). This general histological distinction reflects the clinical and biological behavior of these two tumor types. Approximately half of all non-small cell lung cancers (NSCLC) are either localized or locally advanced at the time of diagnosis and are treated by resection alone, or by combined modality therapy with or without resection. By contrast, small cell lung cancers (SCLC) are metastatic in 80% of cases at diagnosis. The 20% of SCLC that are initially localized to the hemithorax are usually locally advanced tumors managed by combination chemotherapy and radiotherapy. Less than 10% of SCLC are detected at a very early stage when they can be treated by resection and adjuvant chemotherapy.

The TNM staging system has traditionally been used for NSCLC. Although it is supposed to be applied also to SCLC, in practice these tumors have been classified as "limited" or "extensive" disease, a staging system introduced in the 1950s by the Veterans' Administration Lung Study Group for use in their clinical trials. Limited disease (LD) was characterized by tumors confined to one hemithorax, although local extension and ipsilateral supraclavicular nodes could also be present if they could be encompassed in the same radiation portal as the primary tumor. No extrathoracic metastases could be present. All other patients were classified as extensive disease (ED). In 1989, a consensus report from the IASLC recommended that LD be defined as tumors limited to one hemithorax with regional lymph node metastases including hilar, ipsilateral and contralateral mediastinal and ipsilateral and contralateral supraclavicular nodes. This report also recommends that patients with ipsilateral pleural effusion regardless of whether cytology positive or negative should be considered to have LD if no extrathoracic metastases were detected. More recently, analysis of an international database developed by the IASLC that includes 8088 SCLC patients showed that the TNM staging system is applicable to SCLC. Therefore, the staging system being presented in this edition of the staging manual should now be applied to both NSCLC and SCLC.

Bronchopulmonary carcinoid tumors are also frequently classified according to the TNM staging system for NSCLC, even though they are not officially included in the AJCC or UICC staging manuals. Recent analysis of both the SEER and the IASLC international lung tumor databases indicates that the TNM staging system for NSCLC is also applicable to bronchopulmonary carcinoid tumors. Therefore, typical carcinoid and atypical carcinoid tumors should also now be routinely classified according to the TNM system used for NSCLC and SCLC.

Clinical Staging. Clinical classification (cTNM) is based on the evidence acquired before treatment, including physical examination, imaging studies (e.g., computed and positron emission tomography), laboratory tests, and staging procedures such as bronchoscopy or esophagoscopy with ultrasound directed biopsies (EBUS, EUS), mediastinoscopy,

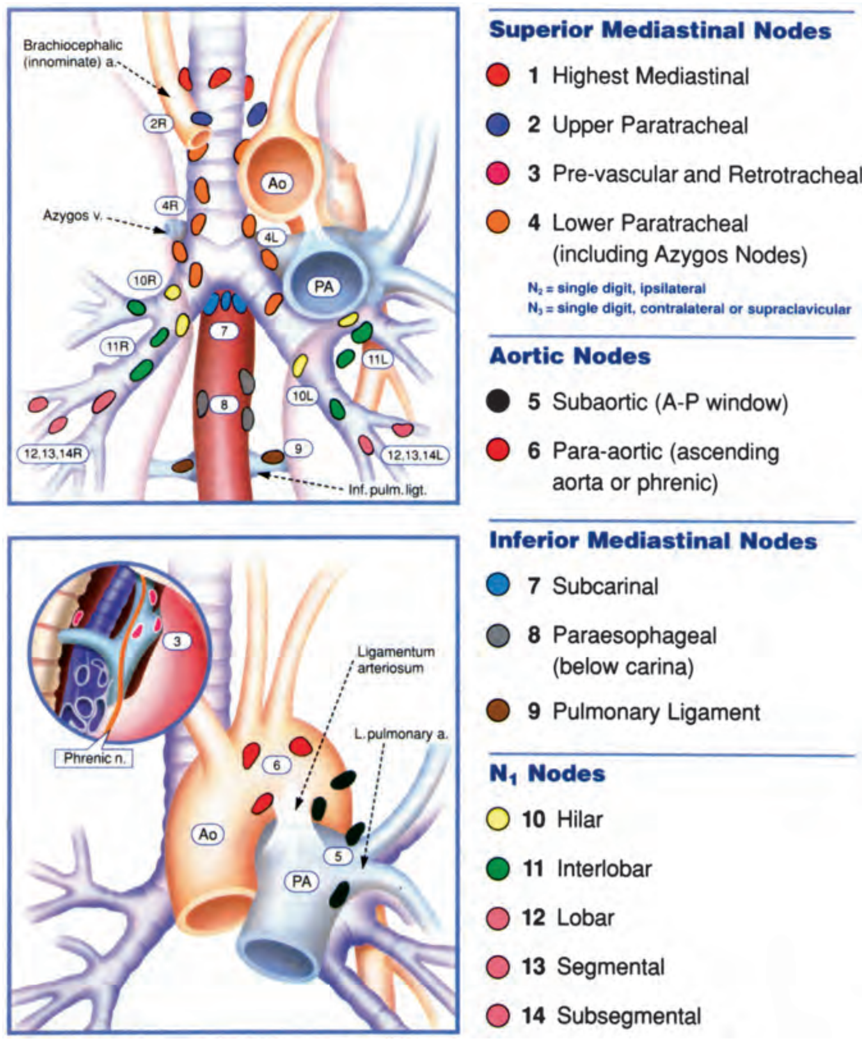

FIGURE 25.2. Mountain/Dresler lymph node map. (From Mountain CF, Dresler CM. Regional lymph node classification for lung cancer staging. Chest 1997;111:1718–1723, used with permission.)

mediastinotomy, thoracentesis, and thoracoscopy (VATS) as well as exploratory thoracotomy.

Pathologic Staging. Pathological classification uses the evidence acquired before treatment, supplemented or modified by the additional evidence acquired during and after surgery, particularly from pathologic examination. The pathologic stage provides additional precise data used for estimating prognosis and calculating end results.

- The pathologic assessment of the primary tumor (pT) entails resection of the primary tumor sufficient to evaluate the highest pT category.
- The complete pathologic assessment of the regional lymph nodes (pN) ideally entails removal of a sufficient number of lymph nodes to evaluate the highest pN category.
- If pathologic assessment of lymph nodes reveals negative nodes but the number of lymph node stations examined are fewer than suggested above, classify the N category as pN0.
- Isolated tumor cells (ITC) are single tumor cells or small clusters of cells not more than 0.2 mm in greatest dimension that are usually detected by immunohistochemistry or molecular methods. Cases with ITC in lymph nodes or at distant sites should be classified as N0 or M0, respectively. The same applies to cases with findings suggestive of tumor cells or their components by non-morphologic techniques such as flow cytometry or DNA analysis.
- The following classification of ITC may be used:

pN0	No regional lymph node metastasis histologically, no examination for ITC
pN0(i−)	No regional lymph node metastasis histologically, negative morphological findings for ITC
pN0(i+)	No regional lymph node metastasis histologically, positive morphological findings for ITC

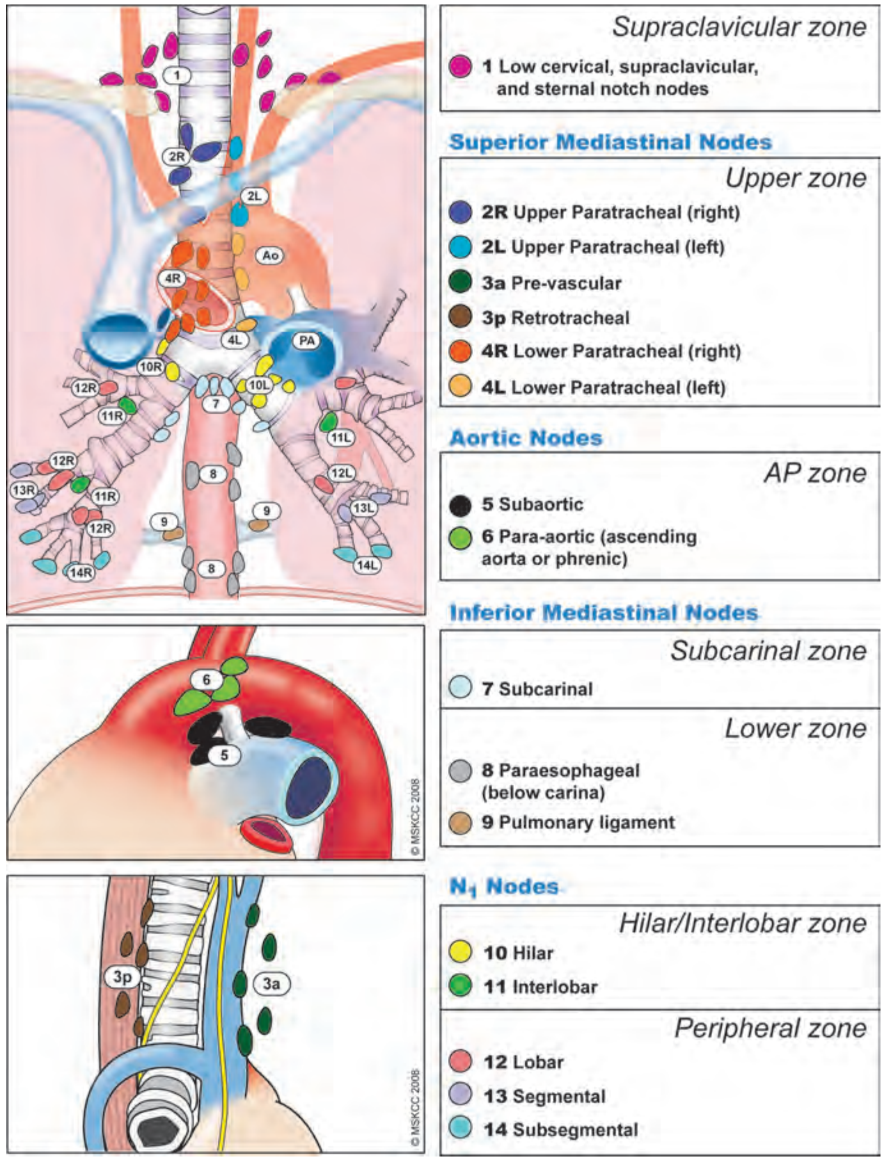

FIGURE 25.3. The IASLC lymph node map shown with the proposed amalgamation of lymph node levels into *zones.* (© Memorial Sloan-Kettering Cancer Center, 2009.)

pN0(mol−) No regional lymph node metastasis histologically, negative non-morphological findings for ITC

pN0(mol+) No regional lymph node metastasis histologically, positive non-morphological findings for ITC

- The pathologic assessment of metastases may be either clinical or pathologic when the T and/or N categories meet the criteria for pathologic staging (pT, pN, cM, or pM).

Pathologic staging depends on the proven anatomic extent of disease, whether or not the primary lesion has been completely removed. If a biopsied primary tumor technically cannot be removed, or when it is unreasonable to remove it, and if the highest T and N categories or the M1 category of the tumor can be confirmed microscopically, the criteria for pathologic classification and staging have been satisfied without total removal of the primary cancer.

Basis for Current Revisions to the Lung Cancer Staging System. The 6th edition of the *AJCC Cancer Staging Manual*, introduced in 2002, made no changes to the previous edition with regards to lung cancer. The proposals for lung cancer staging in the 5th edition, published in 1997, were based on a relatively small database of 5,319 cases of NSCLC accumulated since 1975 by Dr. Clifton Mountain at the MD Anderson Cancer Center (Houston, TX, USA). During this time, there had been many refinements to the techniques available for clinical staging, principally the routine use of computed tomography and more recently, an increasing use of positron emission tomography. The database was largely

TABLE 25.1. Definition for lymph node stations in Japan Lung Cancer Society Map, MD-ATS Map, and IASLC Map

Japan Lung Hancer Society map	*MD-ATS map*	*IASLC map*
	1 Low cervical, supraclavicular and sternal notch nodes	
Located in the area of the upper 1/3 of the intrathoracic trachea. Boundary level from the upper margin of the subclavian artery or the apex to the crossing point of the upper margin of the left brachiocephalic vein and the midline of the trachea	Nodes lying above a horizontal line at the upper rim of the brachiocephalic (left innominate) vein where it ascends to the left, crossing in front of the trachea at its midline	Upper border: lower margin of cricoid cartilage Lower border: clavicles bilaterally and, in the midline, the upper border of the manubrium, 1R designates right-sided nodes, 1 L, left-sided nodes in this region For lymph node station 1, the midline of the trachea serves as the border between 1R and 1 L
2 Paratracheal lymph nodes	*2 Upper paratracheal nodes*	
Located in the area between the superior mediastinal lymph nodes (1) and the tracheobronchial lymph nodes (4). Paratracheal lymph nodes with primary tumor can be defined as ipsilateral lymph nodes; paratracheal lymph nodes without primary tumor can be defined as contralateral lymph nodes	Nodes lying above a horizontal line drawn tangential to the upper margin of the aortic arch and below the inferior boundary of No. 1 nodes	2R: Upper border: apex of the right lung and pleural space, and in the midline, the upper border of the manubrium Lower border: intersection of caudal margin of innominate vein with the trachea As for lymph node station 4R, 2R includes nodes extending to the left lateral border of the trachea 2 L: Upper border: apex of the left lung and pleural space, and in the midline, the upper border of the manubrium Lower border: superior border of the aortic arch
3 Pretracheal lymph nodes	*3 Pre-vascular and retrotracheal nodes*	
Located in the area anterior to the trachea and inferior to the superior mediastinal lymph nodes (1). On the right side, the boundary is limited to the posterior wall of the superior vena cava. On the left side, the boundary is limited to the posterior wall of the brachiocephalic vein 3a Anterior mediastinal lymph nodes On the right side, located in the area anterior to the superior vena cava. On the left side, the boundary is limited to the line connecting the left bracheocephalic vein and the ascending aorta 3p Retrotracheal mediastinal lymph nodes/Posterior mediastinal lymph nodes Located in the retrotracheal or posterior area of the trachea	Prevascular and retrotracheal nodes may be designated 3A and 3P; midline nodes are considered to be ipsilateral	3a: Prevascular On the right: Upper border: apex of chest Lower border: level of carina Anterior border: posterior aspect of sternum Posterior border: anterior border of superior vena cava On the left: Upper border: apex of chest Lower border: level of carina Anterior border: posterior aspect of sternum Posterior border: left carotid artery 3p: Retrotracheal Upper border: apex of chest Lower border: carina

4 Tracheobronchial lymph nodes	*4 Lower paratracheal nodes*	
Located in the area superior to the carina. On the right side, located medial to the azygos vein. On the left side, located in the area surrounded by the medial wall of the aortic arch	The lower paratracheal nodes on the right lie to the right of the midline of the trachea between a horizontal line drawn tangential to the upper margin of the aortic arch and a line extending across the right main bronchus at the upper margin of the upper lobe bronchus, and contained within the mediastinal pleural envelope; the lower paratracheal nodes on the left lie to the left of the midline of the trachea between a horizontal line drawn tangential to the upper margin of the aortic arch and a line extending across the left main bronchus at the level of the upper margin of the left upper lobe bronchus, medial to the ligamentum arteriosum and contained within the mediastinal pleural envelope. Researchers may wish to designate the lower paratracheal nodes as No. 4s (superior) and No. 4i (inferior) subsets for study purposes; the No. 4s nodes may be defined by a horizontal line extending across the trachea and drawn tangential to the cephalic border of the azygos vein; the No. 4i nodes may be defined by the lower boundary of No. 4s and the lower boundary of no. 4, as described above	4R: includes right paratracheal nodes, and pretracheal nodes extending to the left lateral border of trachea Upper border: intersection of caudal margin of innominate vein with the trachea Lower border: lower border of azygos vein 4L: includes nodes to the left of the left lateral border of the trachea, medial to the ligamentum arteriosum Upper border: upper margin of the aortic arch Lower border: upper rim of the left main pulmonary artery
5 Subaortic lymph nodes/Botallo's lymph nodes	*5 Subaortic (aorto-pulmonary window)*	
Located in the area adjacent to the ligamentum arteriosum (Botallo's ligament). The boundary extends from the aortic arch to the left main pulmonary artery	Subaortic nodes are lateral to the ligamentum arteriosum or the aorta or left pulmonary artery and proximal to the first branch of the left pulmonary artery and lie within the mediastinal pleural envelope	Subaortic lymph nodes lateral to the ligamentum arteriosum Upper border: the lower border of the aortic arch Lower border: upper rim of the left main pulmonary artery
	6 Para-aortic nodes (ascending aorta or phrenic)	
Located along the ascending aorta, and in the area of the lateral wall of the aortic arch. Posterior boundary limited to the site of the vagal nerve	Nodes lying anterior and lateral to the ascending aorta and the aortic arch or the innominate artery, beneath a line tangential the upper margin of the aortic arch	Lymph nodes anterior and lateral to the ascending aorta and aortic arch Upper border: a line tangential to the upper border of the aortic arch Lower border: the lower border of the aortic arch
	7 Subcarinal nodes	
Located in the area below the carina, where the trachea bifurcates to the two main bronchi	Nodes lying caudal to the carina of the trachea, but not associated with the lower lobe bronchi or arteries within the lung	Upper border: the carina of the trachea Lower border: the upper border of the lower lobe bronchus on the left; the lower border of the bronchus intermedius on the right
	8 Para-esophageal nodes (below carina)	
Located below the subcarinal lymph nodes, and along the esophagus	Nodes lying adjacent to the wall of the esophagus and to the right or left of the midline, excluding subcarinal nodes	Nodes lying adjacent to the wall of the esophagus and to the right or left of the midline, excluding subcarinal nodes Upper border: the upper border of the lower lobe bronchus on the left; the lower border of the bronchus intermedius on the right Lower border: the diaphragm

continued

TABLE 25.1. Definition for lymph node stations in Japan Lung Cancer Society Map, MD-ATS Map, and IASLC Map (continued)

Japan Lung Cancer Society map	*MD-ATS map*	*IASLC map*
	9 Pulmonary ligament nodes	
Located in the area of the posterior and the lower edge of the inferior pulmonary vein	Nodes lying within the pulmonary ligament, including those in the posterior wall, and lower part of the inferior pulmonary vein	Nodes lying within the pulmonary ligament Upper border: the inferior pulmonary vein Lower border: the diaphragm
	10 Hilar nodes	
Located around the right and left main bronchi	The proximal lobar nodes, distal to the mediastinal pleural eflection and the nodes adjacent to the bronchus intermedius on the right; radiographically, the hilar shadow may be created by enlargement of both hilar and interlobar nodes	Includes nodes immediately adjacent to the mainstem bronchus and hilar vessels including the proximal portions of the pulmonary veins and main pulmonary artery Upper border: the lower rim of the azygos vein on the right; upper rim of the pulmonary artery on the left Lower border: interlobar region bilaterally
	11 Interlobar nodes	
Located between the lobar bronchi. On the right side, subclassified into two groups: 11s: Superior interlobar nodes: located at the bifurcation of the upper and middle lobar bronchi 11i: Inferior interlobar nodes: located at the bifurcation of the middle and lower lobar bronchi	Nodes lying between the lobar bronchi	Between the origin of the lobar bronchi 11s: between the upper lobe bronchus and bronchus intermedius on the right 11i: between the middle and lower lobe bronchi on the right
	12 Lobar nodes	
Located in the area around the lobar branches, which are subclassified into three groups: 12u: Upper lobar lymph nodes 12 m: Middle lobar lymph nodes 12 *l*: Lower lobar lymph nodes	Nodes adjacent to the distal lobar bronchi	Adjacent to the lobar bronchi
	13 Segmental nodes	
Located along the segmental branches	Nodes adjacent to the segmental bronchi	Adjacent to the segmental bronchi
	14 Subsegmental nodes	
Located along the subsegmental branches	Nodes around the subsegmental bronchi	Adjacent to the subsegmental bronchi

from a single institution, containing cases predominantly treated surgically. Repeated iterations of the TNM staging system had seen recommendations for lung cancer staging evolve with little internal validation and no external validation of the descriptors or the stage groupings. Increasingly reports from other databases challenged some of the descriptors and stage groupings. In preparation for this 7th edition of the staging manual, the IASLC established a Lung Cancer Staging Project in 1998 to bring together the large databases available worldwide to inform recommendations for revision that would be intensively validated. The results of this project were accepted by the International Union Against Cancer (UICC) and the AJCC as the primary source for revisions of the lung cancer staging system in the 7th editions of their staging manuals.

The IASLC lung cancer database includes cases from 46 sources in more than 19 countries, diagnosed between 1990 and 2000 and treated by all modalities of care. A total of 100,869 cases were submitted to the data center at Cancer Research and Biostatistics (Seattle, WA, USA). After an initial sift to exclude cases outside the study period, those for whom cell type was not known, cases not newly diagnosed at the point of entry, and those with inadequate information on stage, treatment, or follow-up, 81,015 cases remained for analysis. Of these, 67,725 were NSCLC and 13,290 were SCLC. The analyses of the T, N, and M descriptors and the subsequent analysis of TNM subsets and stage groupings were derived from the cases of NSCLC and subsequently validated also in the SCLC cases and carcinoids. Survival was measured from the date of entry (date of diagnosis for registries, date of registration for protocols) for clinically staged data and the date of surgery for pathologically staged data and was calculated by the Kaplan–Meier method. Prognostic groups were assessed by Cox regression analysis.

Where the analyses showed descriptors to have a prognosis that differed from the other descriptors in any T or M category, two alternative strategies were considered: (1) Retain that descriptor in the existing category, identified by alphabetical subscripts. For example, additional pulmonary nodules in the lobe of the primary, considered to be T4 in the 6th edition, would become T4a, whereas additional pulmonary nodules in other ipsilateral lobes, designed as M1 in the 6th edition, would become M1a. (2) Allow descriptors to move between categories, to a category containing other descriptors with a similar prognosis, e.g., additional pulmonary nodules in the lobe of the primary would move from T4 to T3, and additional pulmonary nodules in other ipsilateral lobes would move from M1 to T4. The first strategy had the advantage of allowing, to a large extent, retrograde compatibility with existing databases. Unfortunately, this generated a large number of descriptors (approximately 20) and an impractically large number of TNM subsets (>180). For this reason, backward compatibility was compromised and strategy (2) was preferred for its clinical utility. A small number of candidate stage grouping schemes were developed initially using a recursive partitioning and amalgamation algorithm. The analysis grouped cases based on best stage (pathologic, if available, otherwise clinical) after determination of best-split points based on overall survival

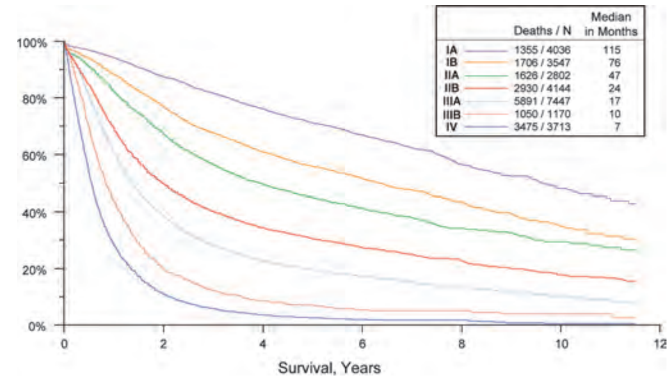

FIGURE 25.4. Survival in all NSCLC by TNM stage (according to "best" based on a combination of clinical and pathologic staging.

on indicator variables for the newly presented TM categories and an ordered variable for N category, excluding NX cases (Figures 25.4 and 25.5). The analysis was performed on a randomly selected training set comprising two-thirds of the available data that met the requirements for conversion to newly presented T and M categories, reserving the other one-third of cases for later validation. The random selection process was stratified by type of database submission and time period of case entry (1990–1994 vs. 1995–2000).

Selection of a final stage grouping proposal from among the candidate schemes was done based on its statistical properties in the training set and its relevance to clinical practice and by consensus.

Table 25.2 shows a comparison of the 6th edition and 7th edition TNM for lung cancer to assure clarity for the user. The final 7th edition TNM is described in the "Definitions of TNM" section that follows.

PROGNOSTIC FEATURES

The IASLC lung cancer database, although retrospective, provides the largest published analyses of prognostic factors in both NSCLC and SCLC. Potentially useful prognostic variables for lung cancer survival that were considered included: TNM

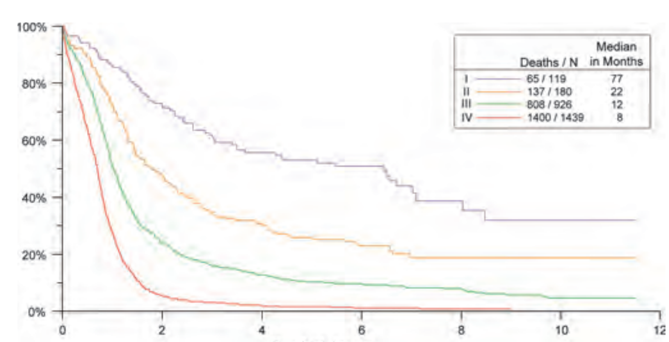

FIGURE 25.5. Survival in all SCLC by TNM stage (according to "best" stage based on a combination of clinical and pathologic staging in the IASLC lung database).

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TABLE 25.2. Stage grouping comparisons: 6th edition vs. 7th edition descriptors, T and M categories, and stage groupings

Sixth edition T/M descriptor	*7th edition T/M*	*N0*	*N1*	*N2*	*N3*
T1 (≤2 cm)	T1a	IA	IIA	IIIA	IIIB
T1 (>2–3 cm)	T1b	IA	IIA	IIIA	IIIB
T2 (≤5 cm)	T2a	IB	**IIA**	IIIA	IIIB
T2 (>5–7 cm)	T2b	**IIA**	IIB	IIIA	IIIB
T2 (>7 cm)	T3	**IIB**	**IIIA**	IIIA	IIIB
T3 invasion	T3	IIB	IIIA	IIIA	IIIB
T4 (same lobe nodules)	T3	**IIB**	**IIIA**	**IIIA**	IIIB
T4 (extension)	T4	**IIIA**	**IIIA**	IIIB	IIIB
M1 (ipsilateral lung)	T4	**IIIA**	**IIIA**	**IIIB**	**IIIB**
T4 (pleural effusion)	M1a	**IV**	**IV**	**IV**	**IV**
M1 (contralateral lung)	M1a	IV	IV	IV	IV
M1 (distant)	M1b	IV	IV	IV	IV

Cells in bold indicate a change from the 6th edition for a particular TNM category.

From Goldstraw P, Crowley J, Chansky K et al: The IASLC Lung Cancer Staging Project: Proposals for the revision of the TNM stage groupings in the forthcoming (seventh) edition of the TNM classification of malignant tumours. J Thorac Oncol 2:706–714, 2007, with permission.

stage, tumor histology, patient age, sex, and performance status, various laboratory values and molecular markers.

Clinical Factors. Analyses of the IASLC lung cancer database revealed that in addition to clinical stage, performance status and patient age and sex (male gender being associated with a worse survival) were important prognostic factors for both NSCLC and SCLC. In NSCLC, squamous cell carcinoma was associated with a better prognosis for patients with Stage III disease but not in other tumor stages. In advanced NSCLC (Stages IIIB/IV), some laboratory tests (principally white blood cells and hypercalcemia) were also important prognostic variables. In SCLC, albumin was an independent biological factor. Analyses that incorporate these factors along with overall TNM stage stratify both NSCLC and SCLC patients into 4 groups that have distinctly different overall survivals. In addition to these, a recent study of 455 patients with completely resected pathologic Stage I NSCLC suggests that high preoperative serum carcinoembryonic antigen (CEA) levels identify patients who have a poor prognosis, especially if those levels also remain elevated postoperatively. Other retrospective studies report that the intensity of hypermetabolism on FDG-PET scan is correlated with outcome in NSCLC patients managed surgically. Additional prospective studies are needed to validate these findings and to determine whether FDG-PET is prognostic across all lung cancer stages and histologies.

In the lung, arterioles are frequently invaded by cancers. For this reason, the V classification is applicable to indicate vascular invasion, whether venous or arteriolar.

Biological Factors. In recent years, multiple biological and molecular markers have been found to have prognostic value for survival in lung cancer, particularly NSCLC. These are summarized in Table 25.3. Although some molecular abnormalities, for example EGFR and K-ras mutations, are now being used to stratify patients for treatment, none is yet routinely used for lung cancer staging.

TABLE 25.3. Metaanalyses published on the prognostic value of biological or genetic markers for survival in lung cancer

Biological variable	*Prognostic factor*	*Reference*
bcl-2	Favorable	Martin et al. 2003
TTF1	Adverse	Berghmans et al. 2006
Cox2	Adverse	Mascaux et al. 2006
EGFR overexpression	Adverse	Nakamura et al. 2006 Meert et al. 2002
EGFR mutation	Favorable	Marks et al. 2007
ras	Adverse	Mascaux et al. 2006 Huncharek et al. 1999
Ki67	Adverse	Martin et al. 2004
HER2	Adverse	Meert et al. 2003 Nakamura et al. 2005
VEGF	Adverse	Delmotte et al. 2002
Microvascular density	Adverse	Meert et al. 2002
p53	Adverse	Steels et al. 2001 Mitsudomi et al. 2000 Huncharek et al. 2000
Aneuploidy	Adverse	Choma et al. 2001

Adapted from Sculier JP et al. The IASLC Lung Cancer Staging Project: The impact of additional prognostic factors on survival and their relationship with the anatomical extent of disease expressed by the 6th edition of the TNM classification of malignant tumours and the proposals for the 7th edition, *J Thorac Oncol* 3(4):457–466, 2008, with permission.

DEFINITIONS OF TNM

Primary Tumor (T)

TX	Primary tumor cannot be assessed, or tumor proven by the presence of malignant cells in sputum or bronchial washings but not visualized by imaging or bronchoscopy
T0	No evidence of primary tumor
Tis	Carcinoma in situ
T1	Tumor 3 cm or less in greatest dimension, surrounded by lung or visceral pleura, without bronchoscopic evidence of invasion more proximal than the lobar bronchus (i.e., not in the main bronchus)*
T1a	Tumor 2 cm or less in greatest dimension
T1b	Tumor more than 2 cm but 3 cm or less in greatest dimension
T2	Tumor more than 3 cm but 7 cm or less or tumor with any of the following features (T2 tumors with these features are classified T2a if 5 cm or less); Involves main bronchus, 2 cm or more distal to the carina; Invades visceral pleura (PL1 or PL2); Associated with atelectasis or obstructive pneumonitis that extends to the hilar region but does not involve the entire lung
T2a	Tumor more than 3 cm but 5 cm or less in greatest dimension
T2b	Tumor more than 5 cm but 7 cm or less in greatest dimension
T3	Tumor more than 7 cm or one that directly invades any of the following: parietal pleural (PL3), chest wall (including superior sulcus tumors), diaphragm, phrenic nerve, mediastinal pleura, parietal pericardium; or tumor in the main bronchus (less than 2 cm distal to the carina* but without involvement of the carina; or associated atelectasis or obstructive pneumonitis of the entire lung or separate tumor nodule(s) in the same lobe
T4	Tumor of any size that invades any of the following: mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, carina, separate tumor nodule(s) in a different ipsilateral lobe

*The uncommon superficial spreading tumor of any size with its invasive component limited to the bronchial wall, which may extend proximally to the main bronchus, is also classified as T1a.

Regional Lymph Nodes (N)

NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastases
N1	Metastasis in ipsilateral peribronchial and/or ipsilateral hilar lymph nodes and intrapulmonary nodes, including involvement by direct extension
N2	Metastasis in ipsilateral mediastinal and/or subcarinal lymph node(s)
N3	Metastasis in contralateral mediastinal, contralateral hilar, ipsilateral or contralateral scalene, or supraclavicular lymph node(s)

Distant Metastasis (M)

M0	No distant metastasis
M1	Distant metastasis
M1a	Separate tumor nodule(s) in a contralateral lobe tumor with pleural nodules or malignant pleural (or pericardial) effusion*
M1b	Distant metastasis (in extrathoracic organs)

From Goldstraw P, Crowley J, Chansky K, et al.: The IASLC Lung Cancer Staging Project: Proposals for the revision of the TNM stage groupings in the forthcoming (seventh) edition of the TNM classification of malignant tumours. *J Thorac Oncol* 2:706–714, 2007, with permission.

*Most pleural (and pericardial) effusions with lung cancer are due to tumor. In a few patients, however, multiple cytopathologic examinations of pleural (pericardial) fluid are negative for tumor, and the fluid is nonbloody and is not an exudate. Where these elements and clinical judgment dictate that the effusion is not related to the tumor, the effusion should be excluded as a staging element and the patient should be classified as M0.

ANATOMIC STAGE/PROGNOSTIC GROUPS

Occult carcinoma	TX	N0	M0
Stage 0	Tis	N0	M0
Stage IA	T1a	N0	M0
	T1b	N0	M0
Stage IB	T2a	N0	M0
Stage IIA	T2b	N0	M0
	T1a	N1	M0
	T1b	N1	M0
	T2a	N1	M0
Stage IIB	T2b	N1	M0
	T3	N0	M0
Stage IIIA	T1a	N2	M0
	T1b	N2	M0
	T2a	N2	M0
	T2b	N2	M0
	T3	N1	M0
	T3	N2	M0
	T4	N0	M0
	T4	N1	M0
Stage IIIB	T1a	N3	M0
	T1b	N3	M0
	T2a	N3	M0
	T2b	N3	M0
	T3	N3	M0
	T4	N2	M0
	T4	N3	M0
Stage IV	Any T	Any N	M1a
	Any T	Any N	M1b

Job Name: -- /381449t

PROGNOSTIC FACTORS (SITE-SPECIFIC FACTORS)
(Recommended for Collection)

Required for staging	None
Clinically significant	Pleural/elastic layer invasion (based on H&E and elastic stains) Separate tumor nodules Vascular invasion – V classification (venous or arteriolar)

HISTOPATHOLOGIC GRADE (G)

GX	Grade cannot be assessed
G1	Well differentiated
G2	Moderately differentiated
G3	Poorly differentiated
G4	Undifferentiated

ADDITIONAL NOTES REGARDING TNM DESCRIPTORS

The T category is defined by the size and extent of the primary tumor. Definitions have changed from the prior edition of TNM. For the T2 category, visceral pleural invasion is defined as invasion to the surface of the visceral pleura or invasion beyond the elastic layer. On the basis of a review of published literature, the IASLC Staging Committee recommends that elastic stains can be used in cases where it is difficult to identify invasion of the elastic layer by hematoxylin and eosin (H&E) stains. A tumor that falls short of completely traversing the elastic layer is defined as PL0. A tumor that extends through the elastic layer is defined as PL1 and one that extends to the surface of the visceral pleural as PL2. Either PL1 or PL2 status allows classification of the primary tumor as T2. Extension of the tumor to the parietal pleura is defined as PL3 and categorizes the primary tumor as T3. Direct tumor invasion into an adjacent ipsilateral lobe (i.e., invasion across a fissure) is classified as T2a. These definitions are illustrated in a report by Travis et al., referenced in the bibliography of this chapter.

Multiple tumors may be considered to be synchronous primaries if they are of different histological cell types. When multiple tumors are of the same cell type, they should only be considered to be synchronous primary tumors if in the opinion of the pathologist, based on features such as associated carcinoma in situ or differences in morphology, immunohistochemistry, and/or molecular studies, they represent differing subtypes of the same histopathological cell type, and also have no evidence of mediastinal nodal metastases or of nodal metastases within a common nodal drainage. Synchronous primary tumors are most commonly encountered when dealing with either bronchioloalveolar carcinomas or adenocarcinomas of mixed subtype with a bronchioloalveolar component. Multiple synchronous primary tumors should be staged separately. The highest T category and stage of disease should be assigned and the multiplicity or the number of tumors should be indicated in parenthesis, e.g., T2(m) or T2(5).

Vocal cord paralysis (resulting from involvement of the recurrent branch of the vagus nerve), superior vena caval obstruction, or compression of the trachea or esophagus may be related to direct extension of the primary tumor or to lymph node involvement. The treatment options and prognosis associated with this direct extension of the primary tumor fall within the T4N0-1 (Stage IIIA) category; therefore, a classification of T4 is recommended. If the primary tumor is peripheral, vocal cord paralysis is usually related to the presence of N2 disease and should be classified as such.

The designation of "Pancoast" tumors relates to the symptom complex or syndrome caused by a tumor arising in the superior sulcus of the lung that involves the inferior branches of the brachial plexus (C8 and/or T1) and, in some cases, the stellate ganglion. Some superior sulcus tumors are more anteriorly located and cause fewer neurological symptoms but encase the subclavian vessels. The extent of disease varies in these tumors, and they should be classified according to the established rules. If there is evidence of invasion of the vertebral body or spinal canal, encasement of the subclavian vessels, or unequivocal involvement of the superior branches of the brachial plexus (C8 or above), the tumor is then classified as T4. If no criteria for T4 disease pertain, the tumor is classified as T3.

Tumors directly invading the diaphragm in the absence of other signs of locally advanced disease are rare, constituting less than 1% of all cases of potentially resectable NSCLC. These tumors are considered to be T3, but appear to have a poor prognosis, even after complete resection and in the absence of N2 disease. The classification of such tumors may need to be reevaluated in the future as more survival data become available.

The term "satellite nodules" was included in the 6th edition of the *AJCC Cancer Staging Manual*. It was defined as additional small nodules in the same lobe as the primary tumor but anatomically distinct from it that could be recognized grossly. Additional small nodules that could be identified only microscopically were not included in this definition. The term "satellite nodules" is being deleted from this edition of the *Staging Manual* because it is confusing, has no scientific basis, and is at variance with the UICC Staging Manual. The term "additional tumor nodules" should be used to describe grossly recognizable multiple carcinomas in the same lobe. Such nodules are classified as T3. This definition does not apply to one grossly detected tumor associated with multiple separate microscopic foci.

HISTOPATHOLOGIC TYPE

The World Health Organization histologic classification of tumors of the lung, 2004, is shown in Table 25.4.

TABLE 25.4. World Health Organization histologic classification of tumors of the lung, 2004

Malignant epithelial tumors	*ICD*
Squamous cell carcinoma	8070/3
Papillary	8052/3
Clear cell	8084/3
Small cell	8073/3
Basaloid	8083/3
Small cell carcinoma	8041/3
Combined small cell carcinoma	8045/3
Adenocarcinoma	8140/3
Adenocarcinoma, mixed subtype	8255/3
Acinar adenocarcinoma	8550/3
Papillary adenocarcinoma	8260/3
Bronchioloalveolar carcinoma	8250/3
Nonmucinous	8252/3
Mucinous	8253/3
Mixed nonmucinous and mucinous or indeterminate	8254/3
Solid adenocarcinoma with mucin production	8230/3
Fetal adenocarcinoma	8333/3
Mucinous ("colloid") carcinoma	8480/3
Mucinous cystadenocarcinoma	8470/3
Signet ring adenocarcinoma	8490/3
Clear cell adenocarcinoma	8310/3
Large cell carcinoma	8012/3
Large cell neuroendocrine carcinoma	8013/3
Combined large cell neuroendocrine carcinoma	8013/3
Basaloid carcinoma	8123/3
Lymphoepithelioma-like carcinoma	8082/3
Clear cell carcinoma	8310/3
Large cell carcinoma with rhabdoid phenotype	8014/3
Adenosquamous carcinoma	8560/3
Sarcomatoid carcinoma	8033/3
Pleomorphic carcinoma	8022/3
Spindle cell carcinoma	8032/3
Giant cell carcinoma	8031/3
Carcinosarcoma	8980/3
Pulmonary blastoma	8972/3
Carcinoid tumor	8240/3
Typical carcinoid	8240/3
Atypical carcinoid	8249/3
Salivary gland tumors	
Mucoepidermoid carcinoma	8430/3
Adenoid cystic carcinoma	8200/2
Epithelial-myoepithelial carcinoma	8562/3

Morphology code of the International Classification of Diseases for Oncology (ICD-0) and the Systematized Nomenclature of Medicine (http://snowmed.org). Behavior is coded /0 for benign tumors, /3 for malignant tumors, and /1 for borderline or uncertain behavior.

From Travis WD, et al., eds. Tumours of the Lung, Pleura, Thymus and Heart. World Health Organization Classification of Tumours. Lyon: IARC Press, 2004, p. 10, with permission of IARC Press.

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